

NAYDA FARNSWORTH

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EDUCATION

Rutgers University

New Brunswick, NJ

Ph.D. in Mathematics

August 2026 – Present

- School of Graduate Studies Fellowship for Excellence Recipient

Colgate University

Hamilton, NY

B.A. in Mathematics and Computer Science, summa cum laude

May 2026

- GPA: 3.94/4.00
- High Honors in Mathematics; High Honors in Computer Science
- Selected Coursework: Combinatorics, Graph Theory, Probability, Ramsey Theory, Theory of Computing

PUBLICATIONS AND PREPRINTS

1. N. Conlon, **N. Farnsworth**, and A. Robertson. *On Hales–Jewett and Related Numbers*. Submitted, 2026. arXiv:2607.18111.
2. P. Breiding, J. Cobb, A. Englander, **N. Farnsworth**, J. Hauenstein, O. Henriksson, D. Johnson, J. Lopez Garcia, and D. Mundayur. *Elimination Without Eliminating: Computing Complements of Real Hypersurfaces Using Pseudo-Witness Sets*. Submitted, 2026. arXiv:2601.04383.
3. H. Kvinge, A. Aguilar, **N. Farnsworth**, G. O’Brien, R. Jasper, S. Scullen, and H. Jenne. *Do Neural Networks Learn Algebraic Worlds? An Investigation into the Group-theoretic Structures Learned by Narrow Models Trained to Predict Group Operations*. Accepted to TAG-DS 2025. arXiv:2601.21150.

TALKS & PRESENTATIONS

A CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry

San José, CR

Presenter, Mini-talk.

08/2026

Title: *On the Hales–Jewett Theorem*

A CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry

San José, CR

Presenter, Poster Presentation.

07/2026

Title: *Computing Improved Bounds on Hales–Jewett Theorem*

Colgate University Mathematics Department Seminar

Hamilton, NY

Presenter, Oral Presentation.

04/2026

Title: *Computers Meet Combinatorics: SAT Solvers and the Search for Hales–Jewett Bounds*

Colgate University Mathematics Department Seminar

Hamilton, NY

Presenter, Oral Presentation.

04/2026

Title: *Computers Meet Combinatorics: SAT Solvers and the Search for Hales–Jewett Bounds*

Rutgers Experimental Mathematics Seminar

New Brunswick, NJ (Virtual)

Presenter, Oral Presentation.

03/2026

Title: *A Computational Approach to Improving Bounds on the Hales–Jewett Numbers*

Nebraska Conference for Undergraduate Wisdom in Mathematics

Lincoln, NE

Presenter, Oral Presentation.

01/2026

Title: *Establishing an Improved Lower Bound on $HJ(3;3)$*

Osaka University Student Research Symposium	Osaka, JP
Presenter, Oral Presentation.	01/2025
Title: <i>Connections between Tridiagonal Matrices and the Fibonacci Sequence</i>	
Colgate Combinatorics Research Symposium in Tokyo	Tokyo, JP
Presenter, Oral Presentation.	01/2025
Title: <i>Connections between Tridiagonal Matrices and the Fibonacci Sequence</i>	
Nebraska Conference for Undergraduate Women in Mathematics	Lincoln, NE
Presenter, Oral Presentation.	01/2025
Title: <i>Connections between Tridiagonal Matrices and the Fibonacci Sequence</i>	
New York Six Undergraduate Research Conference	Clinton, NY
Presenter, Poster Presentation.	11/2024
Title: <i>Optimizing Time to Fairness in Federated Learning on Resource-Constrained Devices</i>	
University of Massachusetts Summer Undergraduate Research Symposium	Amherst, MA
Presenter, Poster Presentation.	08/2024
Title: <i>Optimizing Time to Fairness in Federated Learning on Resource-Constrained Devices</i>	
Nebraska Conference for Undergraduate Women in Mathematics	Lincoln, NE
Co-presenter, Poster Presentation.	01/2024
Title: <i>Reconstructing Monomial Orders</i>	
New York Six Undergraduate Research Conference	Geneva, NY
Co-presenter, Oral Presentation.	11/2023
Title: <i>Reconstructing Monomial Orders</i>	
Colgate University Summer Research Poster Symposium	Hamilton, NY
Co-presenter, Poster Presentation.	07/2023
Title: <i>Reconstructing Monomial Orders</i>	

TEACHING EXPERIENCE

Rutgers University	New Brunswick, NJ
Department of Mathematics	Summer 2026
Directed Reading Program (DRP) Mentor	
Colgate University	Hamilton, NY
Center for Learning, Teaching, and Research	2023 – 2026
- Peer Tutor in Mathematics and Computer Science - Mathematics Department Linear Algebra Tutor - Computer Science Department Discrete Structures Tutor; Applied Machine Learning Tutor	

SELECTED AWARDS AND HONORS

Rutgers University SGS Fellowship for Excellence	2026
Rutgers University	
Edwin J. Downie '33 Award for Mathematics	2026
Colgate University Department of Mathematics	
Edward P. Felt '81 Memorial Prize Fund in Computer Science	2026
Colgate University Department of Computer Science	

Laura Sanchis Award for Excellence in Research	2026
Colgate University Department of Computer Science	
Graduate School Access Fund Fellow	2025 – 2026
Colgate University	
George W. Cobb Award	2025
Colgate University	
Empacher-IRCA Scholar-Athlete Award	2024
Intercollegiate Rowing Coaches Association	
Phi Eta Sigma National Honor Society	2023 – 2026
Colgate University	
Wei Ren '04 Endowed Scholar	2023 – 2026
Colgate University	
Dean's Award with Distinction for Academic Excellence	2022 – 2026
Colgate University	

WORKSHOPS & ACTIVITIES

A CIMPA School on Combinatorial Structures in Geometry, Graphs, and Symmetry	San José, CR
Program Participant	07/2026 – 08/2026
11th Lake Michigan Workshop on Combinatorics and Graph Theory	South Bend, IN
Workshop Participant	05/2026
SuperComputing 2025 (SC25)	St. Louis, MO
Short Course Participant: Julia for High-Performance Computing	11/2025
Preliminary Arizona Winter School	Virtual
Course Participant, Introduction to Mathematical Cryptography	09/2025 – 11/2025
GROW Columbia 2025	New York, NY
Conference Participant	09/2025
Workshop on the Applications of Commutative Algebra: Fields Institute	Toronto, ON
Conference Participant, Numerical Algebraic Geometry Group	05/2025

RESEARCH EXPERIENCE

Pacific Northwest National Laboratory	Richland, WA
Research Intern, Math, Stats, & Data Science / Research Computing	Summers 2025 – 2026
Studied mathematical frameworks for mechanistic interpretability in large language models, combinatorial optimization models for nuclear nonproliferation, and high-performance computing support for laboratory cluster operations.	
University of Massachusetts Amherst	Amherst, MA
NSF REU Participant, Center for Equitable Energy Transition	Summer 2024
Analyzed algorithmic bias in federated learning, evaluating the computational cost and empirical reliability of bias mitigation methods.	
Colgate University Department of Mathematics	Hamilton, NY
Faculty-Initiated Summer Research Fellow	Summer 2023
Studied reconstructible monomial orders, developing algorithms to analyze matrix structures and reconstructibility criteria.	

LEADERSHIP AND SERVICE

Mathematics Club

2025 - 2026

President, Colgate University

Hamilton Central School Coding Club

2023 - 2026

Club Leader and Volunteer, Colgate University COVE

Society for Industrial and Applied Mathematics

2023 - 2026

Vice President, Colgate University Chapter

NCAA Division I Men's Rowing

2022 - 2025

Varsity Coxswain, Colgate Men's Rowing

TECHNICAL SKILLS

Programming Languages: Bash, C, Java, Julia, Python

Research Computing: Git, LaTeX, Linux-based HPC clusters, Mathematica, MATLAB, SageMath